CSCE 611: Advanced Digital Design:
Fall 2026
Updated 8/19/26

1 Catalog Description

Design techniques for logic systems; emphasis on higher-level CAD tools such as hardware description languages and functional modeling. Prerequisites: CSCE 212

2 Meeting Times

Section	**Lecture**	**Lab**
Section 001	M 3:55-5:10 in Carolina Coliseum 3020A	W 3:55 to 5:10 in Swearingen 3D22
Section 002	M 3:55-5:10 in Carolina Coliseum 3020A	F 3:55 to 5:10 in Swearingen 3D22

3D22 combination: 4-5-2-1-3

3 Electronic Resources

1. Project submission, grade dissemination, up-to-date schedule, downloads, announcements, links to useful information, project descriptions are disseminated via Blackboard, https://blackboard.sc.edu.
2. Lecture recordings: https://youtube.com/c/JasonBakos
3. We will use electronic design software installed in the CSE departmental computer labs. The CSE labs are currently located in Swearingen 1D43 and 3D22. Be sure you can log into the lab computers in 3D22.
4. Urgent course announcements are broadcast on the website and to your university e-mail (@email.sc.edu). Please regularly check (or forward) this e-mail address.
5. Each team must find their own preferred method for collaborating (e.g. GitHub, Dropbox, etc.).

Contact Ryan Austin (rmaustin@cse.sc.edu) for Moodle and lab support.

4 Textbook

Digital Design and Computer Architecture 2/e RISC-V Edition
by David Harris and Sarah Harris
Morgan Kaufmann (Elsevier) Publishers
(Required)

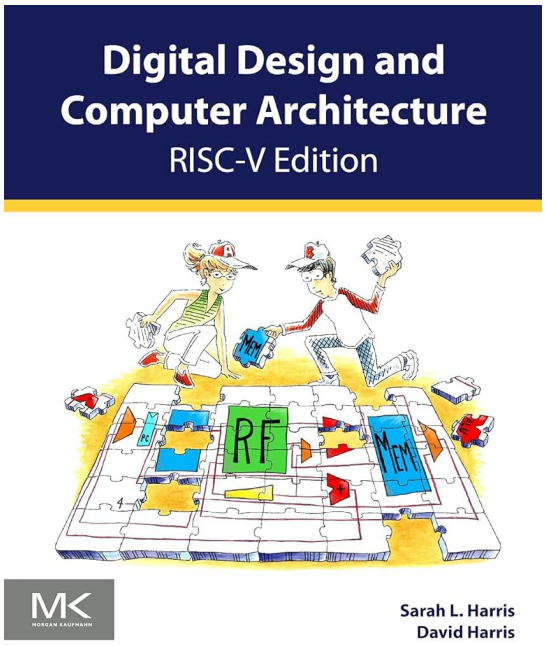


Figure 1: Textbook cover

5 Course Learning Outcomes

- Perform HDL design: Design large-scale digital systems using Verilog
- Perform simulation and verification: Perform behavioral verification using test benches and simulation
- Design microarchitectures: Design a pipelined microprocessor that implements the MIPS instruction set
- Design interconnects: Design a system bus architecture with CPU, memory, and I/O interfaces
- Perform logic synthesis: Synthesize, place-and-route, and implement a computer system on a programmable hardware platform

6 Important Dates

Date	**Day of Week**	**Notes**
Aug. 24	Monday	Last day to change/drop a course without a grade of "W" being recorded
Sept. 7	Monday	Lecture cancelled (Labor Day)
Oct. 16	Friday	Lab cancelled; fall break (sorry Section 002!)
Oct. 19	Monday	In class midterm exam
Nov. 4	Wednesday	Last day to withdraw
Nov. 23, 25, 27	Monday, Wednesday, Friday	Lecture and labs cancelled; Thanksgiving break
Nov. 30	Monday	Last lecture
Dec. 2,4	Wednesday, Friday	Last labs
Dec. 4	Friday	Project submission cutoff
Dec. 7 @ 4:00pm	Monday	In class final exam (150 minutes)

7 Instructors and Teaching Assistants

Instructor:	Jason D. Bakos
E-mail:	jbakos@cse.sc.edu
Office:	M. Bert Storey Engineering and Innovation Center, Room 2263 (note change)
Phone:	777-8627 (x7-8627)
Office hours:	Tuesday, Wednesday, Thursday 11:00 to 1:00
Teaching assistant:	Suyash Singh (Ph.D. student)
E-mail:	ss121@email.sc.edu
Office:	M. Bert Storey Engineering and Innovation Center, Room 2236
Office hours:	by appointment
Teaching assistant:	Alex Anderson-McLeod (Ph.D. student)
E-mail:	aja4@email.sc.edu
Office:	(Will meet in Swearingen 3D22)
Office hours:	by appointment
Teaching assistant:	Junior Tchapmou (undergraduate student)
E-mail:	tchapmou@email.sc.edu
Office:	(Will meet in Swearingen 3D22)
Office hours:	by appointment

8 Grading Structure

Task	**Undergraduate**	**Graduate**
Lab 1: RISCV Program	10%	5%
Lab 2: 7-Segment Decoder	10%	5%
Lab 3: RISCV CPU with R/I/U-type instructions	10%	10%
Lab 4: RISCV CPU with SB/UJ-type instructions	10%	10%
Graduate/Honors Project: VLIW	n/a	10%
Quizzes	15%	15%
Midterm	20%	20%
Final exam	25%	25%
Total	**100%**	**100%**

9 Approximate Schedule

Week	**Lecture**	**Lab**	**Quiz**
8/17	None (semester starts 8/18)	Lecture: Introduction and RISCV ISA	None
8/24	RISCV ISA and SystemVerilog	Lab 1: Square Root	None
8/31	SystemVerilog and Simulation	Lab 1: Square Root	None
9/7	None (Labor Day)	Lab 2: 7-Segment Decoder	Quiz 1
9/14	SystemVerilog Processes (always statement)	Lab 2: 7-Segment Decoder	None
9/21	RISCV Datapath: Arithmetic	Lab 3: RISCV CPU with R/I/U-type Instructions	Quiz 2
9/28	RISCV Datapath: Arithmetic	Lab 3: RISCV CPU with R/I/U-type Instructions	None
10/5	RISCV Datapath: Branch/Jump	Lab 3: RISCV CPU with R/I/U-type Instructions	None
10/12	RISCV Datapath: Branch/Jump	Lab 3: RISCV CPU with R/I/U-type Instructions	None
10/19	Midterm	Lab 4: RISCV CPU with SB/UJ-type Instructions	None
10/26	RISCV Datapath: Branch/Jump	Lab 4: RISCV CPU with SB/UJ-type Instructions	Quiz 3
11/2	Timing Analysis	Lab 4: RISCV CPU with SB/UJ-type Instructions	Quiz 4
11/9	Sequential Logic	Lab 4: RISCV CPU with SB/UJ-type Instructions	Quiz 5
11/16	Digital Building Blocks	Lab 4: RISCV CPU with SB/UJ-type Instructions	Quiz 6
11/23	None (Thanksgiving)	None (Thanksgiving)	None
11/30	CMOS Logic	Lab 4: RISCV CPU with SB/UJ-type Instructions	None

10 Disability Support

The University of South Carolina is committed to providing access to programs and services for qualified students with disabilities. If you are a student with a disability and require accommodation to participate and complete requirements for this class, notify me immediately and contact the Student Disability Resource Center (http://www.sa.sc.edu/sds, 1705 College Street, Close-Hipp, Suite 102, 803-777-6142, sasds@mailbox.sc.edu) for verification of eligibility and determination of specific accommodations. In addition, please provide me the required accommodation letter from the Student Disability Resource Center. All course materials are available in alternative format upon request.

11 Syllabus Change Policy

This syllabus is a guideline for the course and is subject to change with reasonable notice.

12 Grading scale

Grade	Criteria
A	$90 \leq score \leq 100\%$
B+	$85 \leq score < 90\%$
B	$80 \leq score < 85\%$
C+	$75 \leq score < 80\%$
C	$70 \leq score < 75\%$
D+	$65 \leq score < 70\%$
D	$60 \leq score < 65\%$
F	$score < 60\%$

13 Academic Honesty Policy

Students are encouraged to assist their colleagues for the purpose of overcoming technical challenges related to the use of the design tools (foreshadow: like many industrial CAD/EDA tools, the tools we use are non-intuitive, generate meaningless error messages, are unstable, and are poorly documented).

Also, the members of each group are expected to work cooperatively and make nearly equal contributions toward each project.

Any collaboration beyond these exceptions is prohibited and is subject to the university's guidelines, regulations, and policies regarding academic dishonesty.

See the USC Honor Code at bit.ly/USCHonorCode.

Any student caught committing an Honor Code violation will receive a grade of 0 for the corresponding assignment or exam and the instructor will report the violation to the Office of Academic Integrity.

14 Lecture and Lab Attendance Policy

Attendance in lecture and labs is optional but highly encouraged. Lectures will usually appear on Youtube within 24 hours.

Note that the four projects will generally require more than the two, four, or six lab periods allocated to each project (depending on the lab). Significant lab work outside of scheduled lab time may be necessary to complete the projects on time. We suggest that students start each project outside of class time to get the most benefit from TA assistance during the latter part of their project work.

15 Group Work Policy

Group sizes of two are preferred, but group sizes of one are allowed based on enrollment and hardware availability. Three-member groups are not allowed.

Choose your group partner wisely. Both members of each group will receive the same grade for all projects.

16 Project Submissions

- Submitted projects must compile (both hardware and software) to receive partial credit.
- Only one member of each project group need submit each project. Project scores are assigned to each member of the group.
- Each project grade is assigned in part or in whole based on a project demonstration by the project group. All group members must be present for the project presentation.
- Students must submit each project on Dropbox by 11:59PM on the due date to receive full credit.
- Late projects are charged a 10% grade penalty for each school day after the due date, limited to a maximum of 30%. This way, projects submitted more than three days late are still subject to a 30% late penalty. This penalty is deducted from the raw score of the submission, so a one-day late submission earning a raw score of 80% will have a final score of 80% x 90% = 72%.

  Late penalties are based on the date and time at which the project is submitted (uploaded) to Moodle, NOT the date and time of the project demonstration. This way, students may demonstrate a project at any time after project submission without accruing late penalty. However, students must demonstrate the same version of the project that is submitted, and the project will not receive a grade until after the project is demonstrated.
- At most, each group may only submit one version of any project; no resubmissions are allowed.
- Any projects not submitted and demonstrated by 11:59pm on Friday, Dec. 4 will receive a score of 0 without a documented medical excuse.